Bhavya Kapoor

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Summary

MSc Artificial Intelligence & Data Analytics candidate with hands-on experience in Python-based machine learning pipelines, NLP, embedded AI systems, and data analysis. Proficient in end-to-end ML workflows covering preprocessing, model development, and evaluation. Experienced with SQL, Power BI, Tableau, LangChain, RAG, and REST APIs. Comfortable with Git-based development and cross-functional team collaboration.

Education

Amity University, Noida — Noida, India
MSc in Artificial Intelligence & Data Analytics — *2024 – 2026*

- CGPA: 7.43 | Coursework: Machine Learning, Deep Learning, NLP, Statistical Analysis, Data Engineering

Chandigarh University — Chandigarh, India
B.Sc. in Computer Science, Statistics & Mathematics — *2021 – 2024*

- CGPA: 6.46 | Coursework: Data Structures, Python, Database Management, Linear Algebra, Probability & Statistics

Experience

AI/Robotics Research Intern — India
Ingenious Research Solutions Pvt. Ltd.

- Developed an Edge-AI precision irrigation system on ESP32 using lightweight ML models for real-time soil moisture prediction and automated irrigation decisions.
- Designed TinyML inference pipelines and software-based simulations for embedded AI deployment under strict resource constraints.
- Analyzed model optimization strategies including quantization and pruning relevant to TinyML deployments on low-power hardware.
- Authored research documentation and system design analysis for applied Edge-AI prototypes.

Research Engineer Trainee — India
Contrivance Research Solutions Pvt. Ltd.

- Developed embedded control logic for smart home automation devices integrating sensors with microcontroller-based systems.
- Contributed to development and testing of autonomous micro-drone systems for navigation and control validation.
- Validated control workflows and autonomous navigation models for UAV platforms; supported system debugging and performance testing.

Data Science & Machine Learning Intern — India
Infotact Solutions

- Built and optimized end-to-end ML pipelines using Python, Pandas, NumPy, and Scikit-learn for structured and unstructured datasets.
- Developed predictive maintenance models using statistical and ML techniques to forecast equipment failure patterns.
- Contributed to a FinTech NLP document parsing system using Named Entity Recognition (NER) to extract structured entities from financial documents.

Data Science Intern — India
InLighn Tech

- Performed data cleaning, feature engineering, and exploratory analysis using Python, SQL, and Excel across multiple business datasets.
- Built ML models for price prediction, sentiment classification, and multi-class categorization tasks.
- Developed interactive dashboards in Power BI and Tableau to surface actionable business insights for stakeholders.

Product & Growth Associate — India
CalmConnect

- Validated API integrations and backend workflows using Supabase (PostgreSQL); contributed to authentication setup and database configuration.
- Supported product development through manual testing, feature validation, and usability analysis.
- Produced technical documentation for engineering and cross-functional product teams.

R&D Intern – UAV Systems India

Ingenious Research Solutions Pvt. Ltd.

- Implemented PID and Model Predictive Control (MPC) algorithms for UAV flight stabilization on autonomous quadcopter systems.
- Conducted simulation, tuning, and trajectory optimization analysis for autonomous drone navigation.

Projects

ESP32 Edge-AI Irrigation System | *C++, ESP32, TinyML* | GitHub

- Built a real-time automated irrigation controller on ESP32 integrating soil moisture sensors with lightweight ML inference for adaptive watering decisions.
- Designed a software-based simulation pipeline to validate control logic prior to hardware deployment.

Mall Customer Segmentation & Retention Analysis | *Python, Scikit-learn, Pandas* | GitHub

- Applied K-Means Clustering to segment customers into 5 behavioral archetypes based on income and spending patterns; determined optimal cluster count via the Elbow Method and WCSS analysis.
- Built a recency-based retention model to flag at-risk high-value customers and exported a prioritized list for targeted marketing campaigns.

Multimodal Retrieval System | *Python, Gemini API, LangChain, RAG, Streamlit*

- Built a multimodal content retrieval application using Gemini LLM API with LangChain and RAG pipelines supporting both text and image input for contextual search.
- Implemented secure API key management via dotenv and modular application architecture for maintainability.

Disease Prediction Web Application | *Python, Flask, Scikit-learn*

- Developed a Flask-based web application using a Support Vector Classifier (79% accuracy) for multi-class disease prediction.
- Implemented a data preprocessing and feature vectorization pipeline integrated with dataset-driven semantic health recommendations.

Patents (Published – Under Review)

Lead Inventor: Vehicle-to-Driver Pedestrian Warning System App. No. 202511051554

Co-Inventor: Automated System for Detecting Misplaced Books in a Library App. No. 202511052423

Co-Inventor: Enhanced Rain Protection Umbrella App. No. 202511024301

Technical Skills

Languages: Python, SQL, MATLAB, C++ (Basic)
ML & Data Science: Machine Learning, Deep Learning, Predictive Modeling, Feature Engineering, Model Evaluation, Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow
NLP & Generative AI: NLP, Named Entity Recognition (NER), Text Processing, LangChain, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), Generative AI Application Development
Embedded & Edge AI: ESP32, Microcontroller Programming, TinyML, Sensor Integration, PID Control, Model Predictive Control (MPC)
Data Visualization: Power BI, Tableau, Matplotlib, Seaborn
Tools & Platforms: Git, GitHub, Streamlit, Jupyter Notebook, VS Code, REST APIs, Supabase (PostgreSQL), Microsoft Azure (Fundamentals)

Certifications

Google Data Analytics | *Coursera*

Introduction to Artificial Intelligence

Introduction and Application of Python

Data Analysis Fundamentals